UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the May/June 2010 question paper

for the guidance of teachers

0620 CHEMISTRY

0620/32 Paper 32 (Extended Theory), maximum raw mark 80

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1 In **(a)**, **(b)** and **(c)**, descriptions of chemical properties need not be detailed. If more than one answer is given in each section, mark the **first** one and ignore anything subsequent unless it contradicts what they have already written. No marks for reversing physical and chemical properties.

(a) properties should focus on a group 1 metal and not just metals in general

PHYSICAL soft / can be cut (with a knife) / low density / light / low melting point / (good) conductor (heat or electricity) / shiny (when freshly cut) / malleable / ductile / tarnishes [1]

CHEMICAL react with water (**not** steam) / (very) reactive / forms salts with halogens / react vigorously with acids (**ignore** concentration) / forms an alkaline or basic oxide / fixed oxidation state or oxidation number or valency of +1 / has one valency or outer shell electron
not forms ionic compounds on its own. [1]

(b) properties should focus on a transition metal

PHYSICAL hard / high density / dense / high mp or bp / (good) conductor (heat or electricity) / strong / malleable / ductile / silver or grey or lustrous or shiny solid [1]

CHEMICAL more than one oxidation state or valency (**accept** many oxides) / forms coloured compounds or ions (**not** coloured on its own) / forms complex ions / behave as a catalyst / less reactive than group 1 [1]

(c) PHYSICAL colourless gas / yellow gas [1]
not diatomic molecules

CHEMICAL most reactive halogen / **very** reactive / forms **ionic** fluorides / bonds with metals / form **covalent** fluorides / bonds with non-metals / powerful oxidant / gains one electron (to be stable) / fixed oxidation state or valency of –1
allow decolourised when reacts with alkene) / forms F^- ions / forms acidic oxides / forms an acid when reacted with hydrogen / hydride is acidic [1]
not bleaching agent

2 **(a)** **(i)** enzymes are proteins / come from living organisms / biological (catalysts) [1]
not enzymes are living or natural

(ii) carbohydrates have 2H:1O ratio [1]
contain elements of water [1]

contain water = [1]
unless they state that carbohydrates contain water, this response scores 2 or 0

(b) correct -O- linkage [1]
cond same correct monomer (this mark is lost if 2 different boxes are shown) [1]
cond continuation (i.e. bonds at **both** ends) [1]

(c) **(i)** (concentration or amount or mass etc.) of starch decreases (with time) [1]
(concentration etc.) of starch becomes zero / all starch gone [1]
colour (intensity) indicates how much starch is present (can be inferred) [1]

(ii) enzyme denatured / destroyed [1]
not enzymes killed / don't work / saliva denatured

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3 (a) (i) red brown or orange to colourless [1]
not just bromine decolourised
yellow (**not** dark) / white solid / precipitate / goes cloudy [1]
brown to yellow with no mention of solid/precipitate scores = [1]

(ii) $Br_2 + Na_2S \rightarrow 2NaBr + S$ [1]

(iii) look for two comments
sulfide (ion) / sulfur (ion) loses electrons [1]
not sodium sulfide
bromine accepts them [1]

(b) (i) oxidation [1]
not redox

(ii) hydrogen / H_2
not H [1]

(iii) iron(II) hydroxide / ferrous hydroxide [1]

(iv) $4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3$ [1]

(v) oxidation number or state or valency increases / electron loss / Fe^{2+} to Fe^{3+} [1]
not gains oxygen

(vi) sacrificial protection **or** zinc is sacrificed /
zinc corrodes not iron **or** zinc corrodes therefore iron doesn't /
not just zinc rusts
zinc is oxidised in preference to iron /
zinc reacts with oxygen and water in preference to iron /
zinc more reactive or electropositive than iron /
zinc forms ions more readily than iron **or** zinc loses electrons more readily than iron /
electrons move on to iron /
iron is cathode **or** zinc is anode /
any **three** [3]

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4 **(a)** **(i)** same molecular formula / same number of C and H atoms [1]
different structural formula or structure [1]
same compound = [1]

(ii) correct **formula** of but-2-ene / methylpropene / methyl cyclopropane [1]

(iii) bromine / bromine water / aqueous bromine [1]
brown to colourless **not** clear [1]
stays brown [1]
brom**ide** loses the first mark only

OR alkaline potassium manganate(VII) [1]
from purple/pink to green/brown [1]
stays purple [1]

OR acidic potassium manganate(VII) [1]
from purple/pink to colourless **not** clear [1]
stays purple [1]

(b) heat / high temperature (temperature need not be stated, but if it is stated it must be 500ºC or above) [1]

catalyst (need not be named, but if they are named accept any metal oxide or zeolite / aluminosillicates / silicon dioxide) [1]
not nickel/platinum

(c) (1,2)dibromobutane [1]
if numbers given must be correct
butane [1]
butanol [1]
accept butan-1-ol or butan-2-ol **not** but-1-ol / but-1-anol / buthanol

5 **(a)** fractional [1]
distillation [1]

(b) **(i)** O=O / oxygen(–)oxygen / H–H / hydrogen(–)hydrogen [1]

(ii) O-H / oxygen(–)hydrogen / OH / bond between hydrogen and oxygen [1]
not H-O-H

(iii) endothermic. [1]

(c) **(i)** no pollution / no CO / no CO_2 / no oxides of nitrogen / only produces steam or water / no greenhouse gases / no global warming [1]
does not use up fossil fuels / water is not a finite resource / water is a renewable source of energy / hydrogen is renewable / available from electrolysis of water [1]

(ii) obtaining hydrogen from water requires fossil fuels / storage problems / transport problems / limited range of vehicles available / gaseous nature means only produces small amount of energy per unit volume / methane as a source of steam reforming is finite / lack of distribution network [1]
not expensive / anything regarding safety / flammability / explosiveness

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6 (a) (i) Tl_2S [1]

(ii) $TlCl_3$ [1]

(b) filter / centrifuge / decant
wash the precipitate
dry the solid / heat the solid (in oven) / press between filter paper [3]

all three stated but not in correct order = [2]
two out of three stated in any order = [1]

(c) (i) silver chloride / silver bromide [1]
photography / cameras / films / photo chromic lenses / sunglasses [1]

(ii) increase distance between lamp and paper **or** put lamp far away /
put a screen **or** translucent **or** semi-opaque material between them /
use a less powerful **or** low voltage **or** dim lamp /
lower the temperature
any **two** [2]

(d) (i) thalium sulfate + ammonia + water [1]

(ii) $2TlOH + H_2SO_4 \rightarrow Tl_2SO_4 + 2H_2O$ [2]
not balanced = [1]
incorrect formula = [0]

(iii) green precipitate **or** solid (ignore shades of green but not bluey green etc.) [1]
$Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$ **accept** multiples [1]

7 (a) sodium is expensive / difficult to obtain sodium (from sodium chloride) / problems getting electricity / hard to extract sodium / high energy costs in extraction of sodium [1]

(b) (i) reduce temperature / reduce melting point (to 900/1000°C) temperature need not be stated, but if it is stated it must be within the range
better conductivity / solid aluminium oxide does not conduct
aluminium oxide is insoluble in water any **two** [2]

(ii) $2O^{2-} \rightarrow O_2 + 4e^-$ [2] or [0]

(iii) they burn (away) / react with oxygen / form carbon dioxide [1]

(c) hydrogen formed / aluminium above hydrogen in reactivity series / H^+ discharged in preference to Al^{3+} / aluminium is more reactive than hydrogen [1]
aluminium more reactive than carbon / carbon cannot reduce aluminium oxide /
aluminium is higher than carbon in the reactivity series / carbon doesn't reduce aluminium oxide / carbon doesn't displace aluminium [1]
comparison is essential for mark

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8 (a) (i) accept all metals excluding Group I (lithium is acceptable) [1]
not lead **accept** silver

(ii) M nitrite / nitrate(III) [1]
not nitride

(b) (i) exothermic [1]
not reverse reaction is endothermic as the question asks about the forward reaction
cond forward reaction favoured by low temperature / reverse reaction favoured by high temperature [1]
second mark only scores if exothermic is correct.

(ii) position of equilibrium to right / forwards / more products / more N_2O_4 / lighter colour [1]
because this side has smaller volume / fewer moles [1]

(c) if the final answer is between 86–89% award all 4
if the final answer is between 66–67% award 3 marks (M_r of 32 must have been used)
for all other answers marks can be awarded using the mark scheme as below and applying ecf if necessary

number of moles of O_2 formed = 0.16/24 = 0.0067/0.00667 or 1/150
number of moles of $Pb(NO_3)_2$ in the sample = 0.0133/0.013 or 1/75
mass of one mole of $Pb(NO_3)_2$ = 331 g
mass of lead(II) nitrate in the sample = 4.4(1) g
percentage of lead(II) nitrate in sample = 88.3% (**allow** 88–89) [4]

mark **ecf** in this question but **not** to simple integers
if mass of lead(II) nitrate > 5.00 only marks 1 and 2 available
If divides by 32 (not 24) only last 3 marks can score consequentially

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